Meriem Trabelsi

Software Engineer · systems integration and process automation
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Open to relocation within the EU · Arabic (native) · French (C1) · English (B2)

PROFILE

I build systems where the business process is an executed model rather than logic buried in code. My field is the integration of telecom systems aligned with TM Forum standards: BPMN orchestration, event-driven choreography and externalised business rules.

EXPERIENCE

Process Automation Engineer — IT Serv
January 2026 – September 2026 · final-year engineering project, highest honours
Sole designer and developer of an Order-to-Activate orchestration platform for a fibre operator, from address qualification to first invoice. Proof of concept: software layers complete and measured, physical network simulated.

- **Executable BPMN model (118 nodes, five lanes)** run by Kogito and compiled to Java at build time, so a modelling error breaks the build, not a customer order.
- **17 TM Forum Open APIs (v4)** across nine Quarkus microservices, each with its own PostgreSQL database; 13 validated against the official Conformance Test Kits.
- **52 Drools business rules** kept out of the code, so a discount or a threshold changes without a code change.
- **Saga compensation**: a failed journey is undone in reverse order and the fibre port returns to the pool.
- **Event-driven choreography on Apache Kafka**: 23 topics in CloudEvents 1.0, with process instance correlation.
- **GenAI customer assistant**, read-only and answering from the Open APIs, so every figure is real; Kill Bill billing; Angular portal with four role areas.
- **Measured**: 117 unit tests, 10 end-to-end journeys (54 assertions), 50 concurrent orders with no drift in the median. Found and fixed a race condition that gave one fibre port to eight orders.

Java 17 · Quarkus · Kogito · Drools · Apache Kafka · PostgreSQL · Hibernate Reactive · Kill Bill · Angular · Docker · GitHub Actions

Deep Learning Engineer, Computer Vision — Smart Elastic Solutions
June 2025 – July 2025 · internship
SSD MobileNet object detection for an industrial use case. Purpose-built dataset, model tuning and an automated Python pipeline.

Front-End Developer — AllaCarta
July 2024 · internship
Integration of a React template into a production application.

Full-Stack Developer — Archicompass *(French startup)*
February 2023 – September 2023 · internship
Health-compliance application in Spring Boot and Angular. Weekly Scrum sprints, authentication and authorisation through Keycloak.

Web Developer — ADS Valley
June 2022 – July 2022 · internship
Application for sharing internship experiences.

PROJECTS

Exam Invigilation Planner — independent product
May 2025 – October 2026 · github.com/mariem-trabelsi/exam-slot-planner-pro

- **Python desktop app** that assigns teachers to exam slots with a genetic algorithm under twelve configurable rules, with a rule-by-rule compliance report.
- **Mandatory rules enforced by a repair step**: the earlier penalty approach let through 1,485 quota overruns on a real session of 126 invigilators.
- **Evaluation cut from 9.51 ms to 1.39 ms**; 49 automated checks.

EDUCATION

Engineering Degree in Software Engineering (five-year programme, MSc equivalent) — Higher Institute of Computer Science (ISI), University of Tunis El Manar · 2023 – 2026
BSc, Software Engineering and Information Systems — ISI, University of Tunis El Manar · 2020 – 2023

CERTIFICATIONS

Introduction to TM Forum and the Open Digital Architecture — TM Forum, May 2026
DevNet Associate — Cisco, 2025

SKILLS

Languages and frameworks — Java 17, Quarkus, Spring Boot, Hibernate Reactive, Python, Angular, TypeScript, React
Process and integration — BPMN 2.0, Kogito, Drools, Apache Kafka, CloudEvents, REST APIs, microservices, sagas, GenAI
Telecom — TM Forum Open APIs v4, eTOM, SID, ODA, Order-to-Activate
Data and tooling — PostgreSQL, SQL, Docker, Docker Compose, Git, GitHub Actions, Maven, Keycloak, Scrum, load testing